



One consistent burning calendar for anywhere on Earth

Harmonized MODIS + VIIRS active fires, 2000 → today

Challenge Harmonization of MODIS and VIIRS Hot Spots

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Live samuelakosaonyejekwe.github.io/firecal



The problem

Two sensors, one broken record

MODIS (Terra/Aqua, 1 km, 2000 →) and **VIIRS** (S-NPP, 375 m, 2012 →) see the same fires at different resolutions.

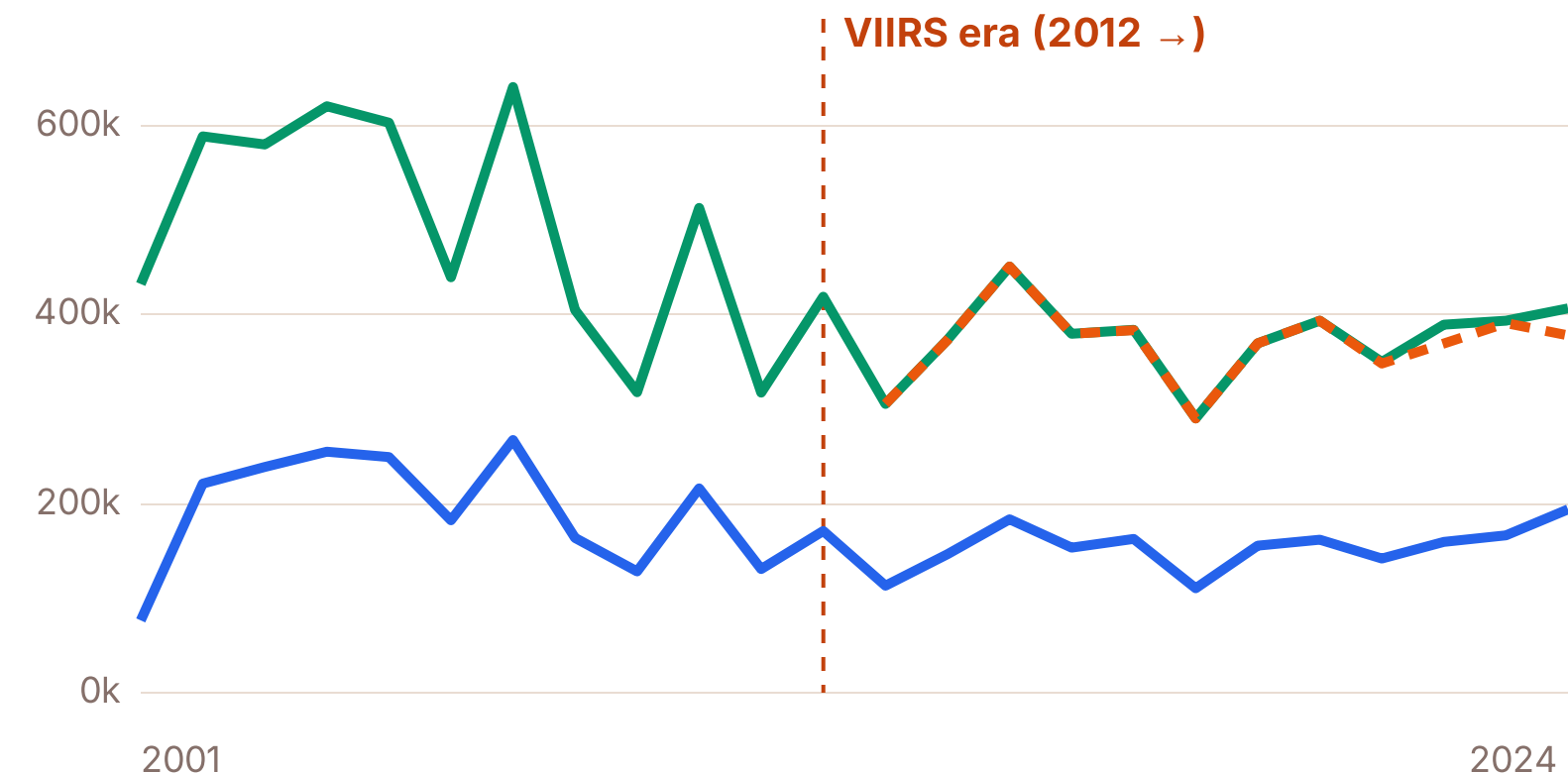
1.9 – 4.0×

more VIIRS fire cell-days than MODIS, varying by region (country ratios)

Raw series jump at the switch: trends, anomalies and seasons become **meaningless** across 2012.

Brazil, annual fire cell-days

— MODIS (raw) — VIIRS (raw) — FireCal harmonized



Harmonized = MODIS rescaled to VIIRS-equivalent before 2012, equal to VIIRS after: no jump.

The method

From hotspots to a harmonized record

1

Clean

Vegetation fires only
(**type 0**), MODIS
conf ≥ 30 , VIIRS
nominal/high, gas
flares masked

2

Grid

0.1° cells $\times$ day $\times$
sensor $\rightarrow$ **fire cell-**
days: removes most
of the 1 km vs 375 m
pixel effect

3

Calibrate

Per area & month on
2012+ overlap,
shrunk to area then
world ratio:

$$k_m = (\Sigma V_m + \lambda \cdot k_{\text{area}}) / (\Sigma M_m + \lambda)$$

4

Three eras

Terra-only (2000–02)
and Terra+Aqua (to
2012) rescaled to
VIIRS-equivalent, as
are VIIRS outage
days; drifting Terra
excluded

5

Validate

Leave-one-year-out
CV + Terra-only vs
Terra+Aqua check;
weak areas flagged
“indicative”

Worldwide prior $k_{\text{world}} = 2.653$ pooled over all 208 countries (fixed, versioned) $\cdot \Lambda = 50, \lambda = 20$

Validation

Accurate where it matters

208

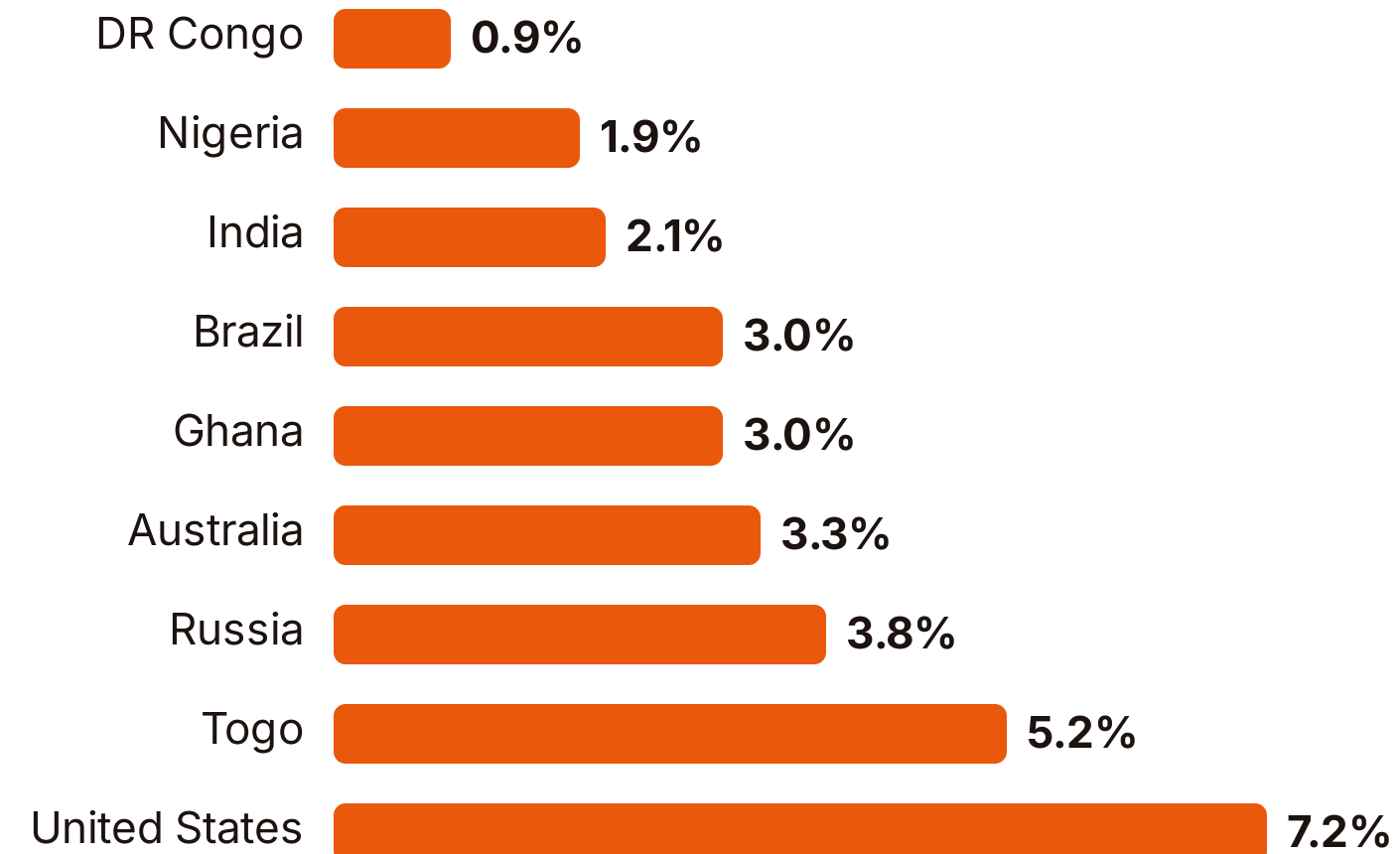
countries harmonized, plus any drawn box

0.9 – 7.2%

median out-of-sample annual error in
major fire regions

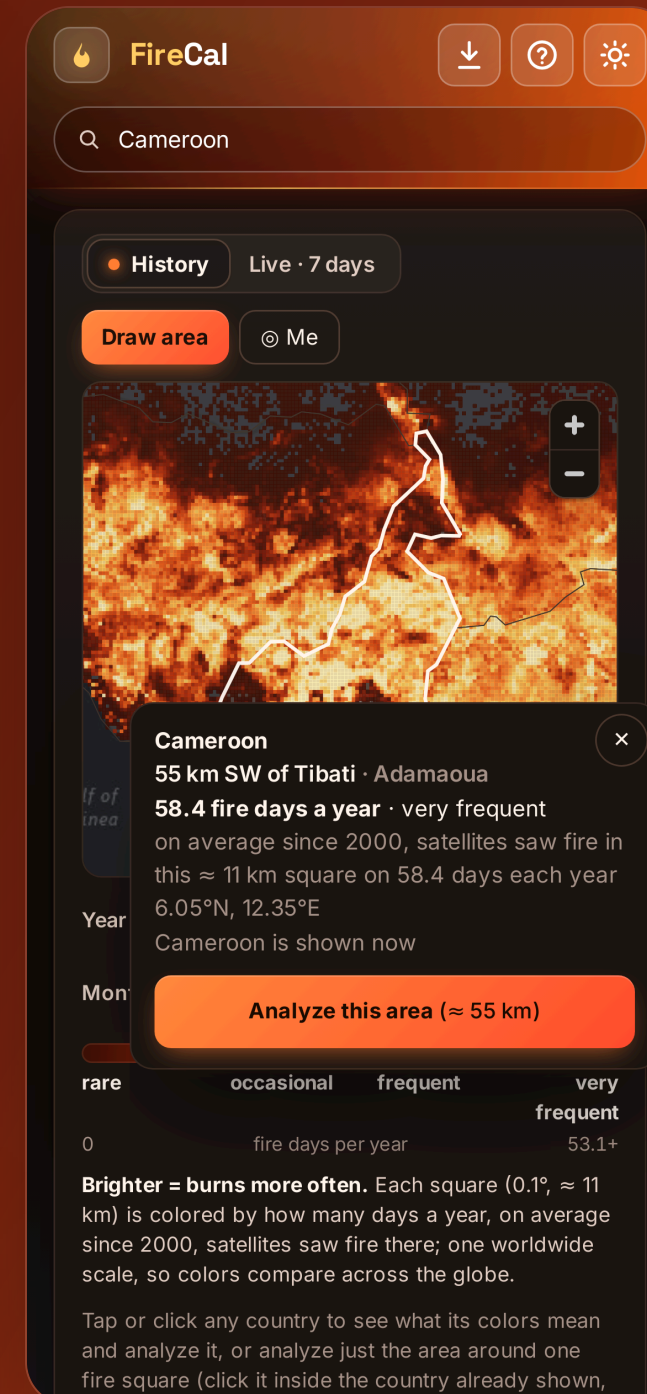
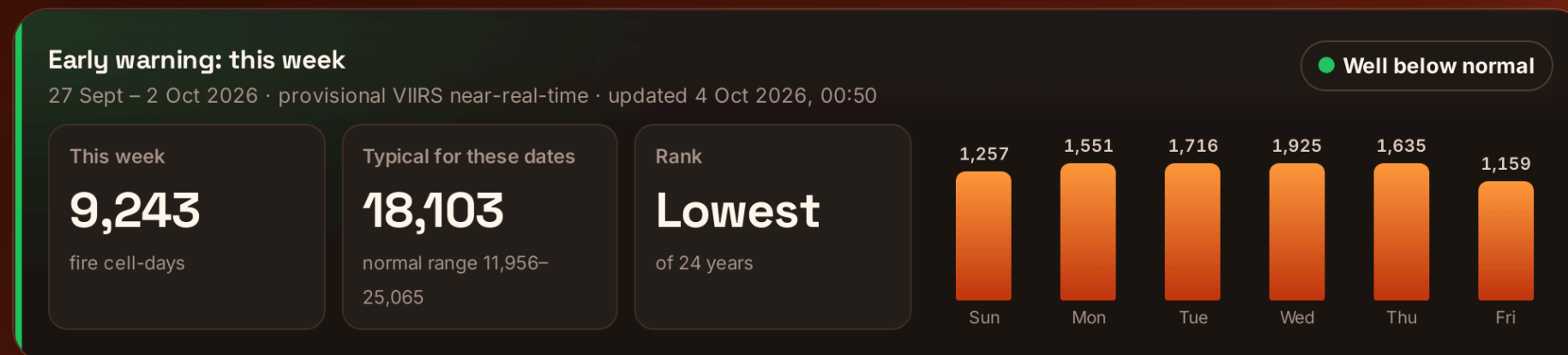
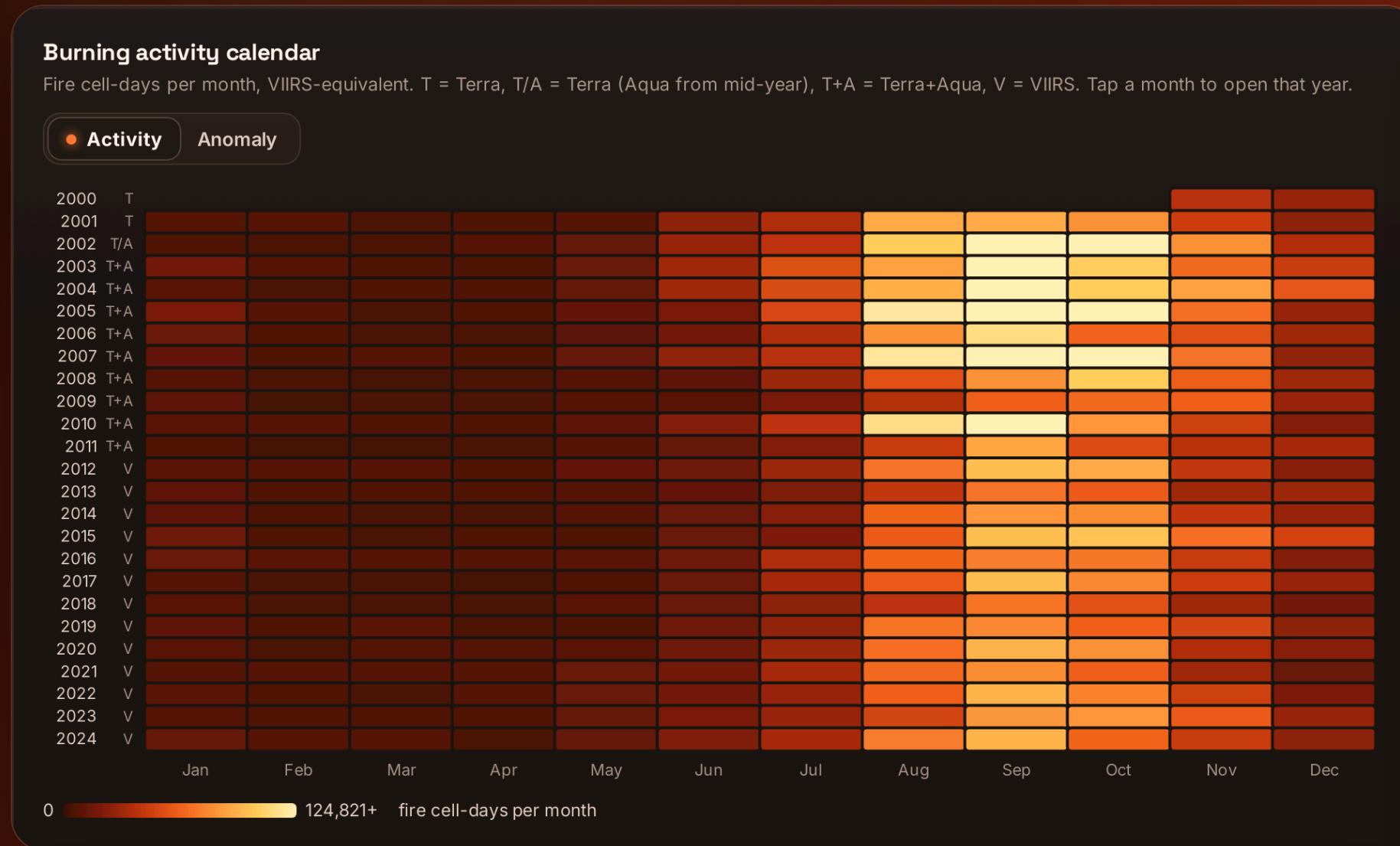
Low-fire areas (e.g. Cyprus, 13.9%) are
flagged **"indicative only"** with the reason.

Leave-one-year-out error (VIIRS totals predicted from MODIS)



The app

Answers in seconds, on any device



Calendar

month × year, anomalies, seasons

Early warning

this week vs every year since 2001

Any area

search, tap, draw, @ locate, or ~55 km round any square

Install & offline

Android, iPhone, desktop

Built on NASA FIRMS open data

MODIS C6.1

Terra/Aqua active fires, FIRMS archive, 2000–2024, all 208 countries

VIIRS S-NPP 375 m

Calibration reference, FIRMS archive 2012–2024

VIIRS NRT (LANCE)

Global 7-day & 24 h feeds for live fires and the nowcast

Pipeline

Python · DuckDB · Parquet;
cloud world build on GitHub
Actions

Two editions

FastAPI server + static site
with an in-browser engine:
identical numbers

Always live

Self-renewing watcher;
browsers read NASA directly
if publishing stalls

Tested

80+ automated tests, parity
checks, accessibility audit
clean

Also: Natural Earth borders (public domain) · GeoNames places (CC BY 4.0) · MapLibre GL JS · Apache ECharts · open source, Apache-2.0

Impact

Know the fire season. Anywhere on Earth.

Responders

Is this week unusual? Which weeks carry the highest risk?

Land managers

Season onset and length trends; quiet windows for fuel management

Scientists

A consistent 2000 → series with transparent calibration and open data

Next

- Add NOAA-20/21 VIIRS for more daily overpasses
- Alerts when a watched area turns unusual
- Link to burned area and emissions products

samuelakosaonyekwe.github.io/firecal

github.com/samuelakosaonyekwe/firecal